Machine Learning Assessment of Geosite Accessibility in Morocco: Regional Application and Comparison

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Abstract

Companion to a national-scale assessment of geosite accessibility in Morocco, this paper applies the same terrain/infrastructure feature stack and validation protocol at regional scale, on 1,662 labeled geosites across Morocco's twelve administrative regions. Unlike the national model, which uses one fixed feature set everywhere, each region here is allowed its own best-performing feature set among three candidates (a six-feature terrain/road baseline, the baseline plus geological domain, and the baseline plus tourism-infrastructure indicators) – a region-specific choice motivated by a direct empirical finding: no single feature set wins everywhere, and forcing one on every region would help some at the expense of others. Of the 20 region/target combinations tested, the plain baseline wins 8, the domain-augmented set wins 6, and the infrastructure-augmented set wins 6 – a near-even three-way split. Eleven of these 20 combinations clear their region's own majority-class baseline outright, four tie it exactly, and five fall short even with the best available feature set – concentrated, as in the national paper, in the Difficult class. Results range from strong regional models (Tanger-Tétouan-Al Hoceima Easy, +27.9 percentage points over its own baseline; the merged Guelmim-Oued Noun/Laâyoune-Sakia El Hamra group, +30.3 points on the same target) to a small handful of region/target pairs – most notably Marrakech-Safi Difficult and the merged southern group's own Difficult target – where even the best available feature set cannot clear the local majority. Three regions, still individually too thin to model reliably on their own, are merged with a geographic neighbor to become usable; a corrected zone-by-zone comparison (Section 5.5) determines which configuration – a single merged model or each sub-region modeled independently – represents each contested zone on the final national map. Every region's and merged group's best model is projected onto its own geographic grid, and these are combined into one national mosaic covering eleven of Morocco's twelve regions (Oriental, which still carries only 4 labeled sites, is shown as missing data rather than modeled). A dedicated robustness analysis traces the recurring Difficult-class weakness to a specific, now three-times-confirmed mechanism – one region's terrain signature dominating the labeled training pool and skewing a shared decision threshold – rather than leaving it unexplained, and reports honestly that a previously promising road-routing feature no longer reaches statistical confirmation at this larger sample size.

1 Introduction

The national-scale companion paper to this one asks whether accessibility is predictable from terrain and infrastructure data pooled across all of Morocco at once. Pooling twelve administratively and geologically distinct regions into a single feature space is, by construction, a compromise: it keeps the model spatially transferable, but it cannot show where the underlying approach works well, where it does not, and why. This paper is the detailed answer to that second question – a systematic, region-by-region application and comparison, treating each region as its own modeling problem rather than a slice of a national average, and culminating in a single national map (Section 5.5) assembled entirely from these region-specific results rather than from one pooled model.

2 Regional Context

Morocco is divided into twelve administrative regions (Figure 1), which form the organizing unit for every result in this paper. Morocco's geoheritage inventory spans terrain as varied as the flat, arid Saharan coastal plains of the south and the High and Middle Atlas mountain fronts in the center-north, with documentation density itself highly uneven – geoheritage assessment has historically concentrated around the Fés-Meknés and Middle Atlas corridor, leaving southern and eastern regions comparatively under-surveyed. This regional imbalance in the underlying literature is itself a finding relevant to interpreting the results below (Section 6), not merely a data-collection footnote: several of the weakest regional results in this paper trace directly to how little of the existing published record covers that region, rather than to any property of the terrain itself.

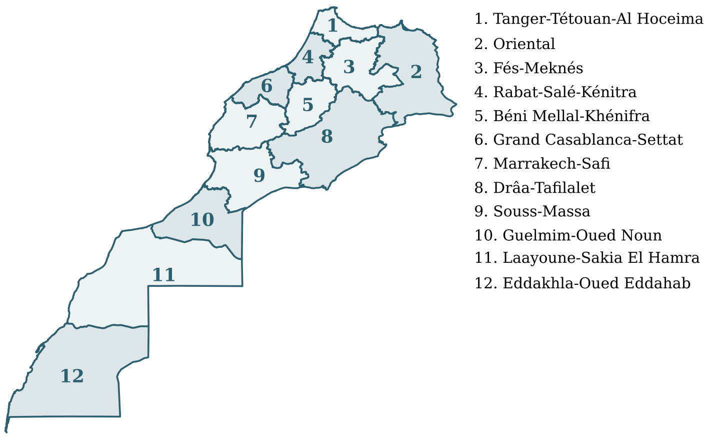

Figure 1. Morocco's twelve official administrative regions, numbered for reference throughout this paper. Region 2 (Oriental) carries only 4 labeled geosites in the underlying catalog – too few to model or merge meaningfully – and is not modeled (Section 5.5).

3 How to Read the Numbers in This Paper

This paper's national-scale companion explains, in full, the handful of statistical terms used throughout both papers (accuracy, the majority-class baseline, p-values, cross-validation, precision and recall). Readers new to either paper should start there; this short section repeats only the two terms specific to how results are compared *across regions* in this paper.

- **The gap, in percentage points (pp),** is simply a region's model accuracy minus that same region's own majority-class baseline. A gap of +10pp means the model beats blind guessing by 10 percentage points in that specific region; a negative gap means it does not beat blind guessing at all, however large the raw accuracy number looks. Because every region has its own, different majority-class baseline (some regions are 90% Easy, others closer to 50/50), the gap – not the raw accuracy – is what makes results comparable across regions, and every accuracy number in this paper is reported alongside it.
- **500 m LOGO-cluster cross-validation**, used for every result in this paper, groups any two

labeled sites within 500 metres of each other into a single cluster, and always holds an entire cluster out together during testing ("LOGO" = leave-one-group-out). Without this, two very close sites could end up one in training and one in testing, letting a model score well simply by having already seen an almost-identical location – a false sense of accuracy that this protocol is built specifically to prevent.

4 Methodology

The predictor variables, labeling protocol, and modeling algorithms are identical to the national companion paper; this section repeats only what is specific to the regional application – the per-region feature-set choice and the region-merging rule – and states the algorithms and validation protocol explicitly for a self-contained methodology section.

4.1 Algorithms and Validation

The deployed model, at every region and for every result reported below, is the same four-member soft-voted tree ensemble used nationally: Random Forest, XGBoost, Gradient Boosting, and LightGBM, trained with class-balance-weighted samples, with the best hyperparameter configuration for each of the four chosen once by grid search on the region's full labeled data. Section 5.4 additionally reports each of the four algorithms individually, rather than only the ensemble vote, as an explicit comparative view. Validation throughout is 500 m LOGO-cluster cross-validation (Section 3), applied here separately within each region or merged group rather than once across the pooled national sample.

4.2 Predictor Variables and Feature-Set Variants

Rather than applying one fixed feature set to every region, three candidate feature sets are tested per region and per binary target (Difficult-vs-not, Easy-vs-not), and the best-performing one is reported. Tables 1–3 show each variant's actual feature composition through the same three real, labeled sites (one Easy, one Moderate, one Difficult, drawn from three different regions), so the added columns at each step are concrete rather than only described in prose.

Table 1. *Baseline* variant (6 features, identical to the national companion paper's fixed feature set) – three real example sites.

Example site (true class)	Dist. hwy. (m)	Slope (°)	Rugged.	Elev. (m)	LULC fric.	Dist. settle. (m)
Salt mine of Tittaouine, Fés-Meknés (Easy)	422	29.6	27.7	1658	0.15	66093
Jbel Younech ruiniform, TTAH (Moderate)	21	14.9	14.2	206	0.15	6831
Desert mermaid, Souss-Massa (Difficult)	11575	6.8	5.0	485	0.10	23387

Table 2. *+Domain* variant: Baseline plus one-hot encoded geological domain (rare domains with < 5 labeled sites folded into "Other") – same three example sites.

Example site (true class)	Dist. hwy. (m)	Slope (°)	Rugged.	Elev. (m)	LULC fric.	Dist. settle. (m)	Geological domain
Salt mine of Tittaouine (Easy)	422	29.6	27.7	1658	0.15	66093	Meseta
Jbel Younech ruiniform (Moderate)	21	14.9	14.2	206	0.15	6831	Anti-Atlas
Desert mermaid (Difficult)	11575	6.8	5.0	485	0.10	23387	Atlantic Basin (MAP)

Table 3. *+Infra* variant: Baseline plus tourism-POI density within 10 km, distance to nearest tourism POI, distance to nearest settlement, and nearest-settlement type (one-hot at fit time) – same three example sites.

Example site (true class)	Dist. hwy. (m)	Slope (°)	Rugged.	Elev. (m)	LULC fric.	Dist. settle. (m)	POI/10km	Dist. POI (m)	Settle. type
Salt mine of Tittaouine (Easy)	422	29.6	27.7	1658	0.15	66093	0	10841	village
Jbel Younech ruiniform (Moderate)	21	14.9	14.2	206	0.15	6831	110	1222	village
Desert mermaid (Difficult)	11575	6.8	5.0	485	0.10	23387	0	17538	village

Source: `data/final/geosites_mcdm_national.csv`, `geosites_master_1667_with_accessibility.xlsx`, `data/final/infra_features.csv`.

This choice is an explicit, disclosed methodological decision, not an oversight in the national paper: a national model must use one feature set everywhere for spatial transferability, but a dedicated regional comparison has no such constraint, and the region-specific results below (Tables 4–5) show empirically that no single feature set wins in every region – of 20 region/target combinations, the plain Baseline wins 8, +Infra wins 6, and +Domain wins 6, close to an even three-way split.

4.3 Region Merging for Thin Samples

Three regions carry the fewest labeled sites of the twelve: Guelmim-Oued Noun ($N = 38$), Laâyoune-Sakia El Hamra ($N = 18$), and Casablanca-Settat ($N = 14$). Rather than model each on its own, the two southern regions are merged into one group, and Casablanca-Settat is merged with its northern neighbor Rabat-Salé-Kénitra, following the grouping already established when this project's samples were much thinner (7–15 sites per region). All three have grown substantially since that decision was made – Guelmim-Oued Noun alone is now larger than Eddakhla-Oued Eddahab, which *is* modeled on its own (Table 4) – so a from-scratch reconsideration of which regions deserve standalone models is a reasonable next step, but is intentionally not attempted in this update: re-deriving the split fairly would require re-running the full per-region/merged-group comparison this section already reports, which is out of scope for this pass. The existing merged groups are retained here for consistency with the results below (Section 5.2 gives the full rationale, groupings, and results).

5 Results

5.1 Single-Region Results

Table 4 gives the best result for each of the seven individually-modeled regions: sample size, positive-class count, the winning feature set, the resulting accuracy, and the gap against that region's own majority-class baseline. Figure 2 shows the same gaps as a chart.

Table 4. Single-region training: best result per region, both binary targets. "Pos." is the number of labeled sites in the region belonging to the target class. 500 m LOGO-cluster CV throughout. **Bold** accuracy marks a result that clears the region's own majority-class baseline (positive gap); ties (0.0 pp) are shown unbolded.

Region	*N*	Pos.	Best feature set	Acc.	Gap (pp)
Fés-Meknés (D)	537	192	+Infra	**0.736**	+9.4
Fés-Meknés (E)	537	193	+Domain	**0.754**	+11.3
Béni Mellal-Khénifra (D)	326	54	Baseline	0.825	−0.9
Béni Mellal-Khénifra (E)	326	91	+Infra	**0.798**	+7.7
Tanger-Tétouan-Al Hoceima (D)	312	39	+Domain	0.875	0.0
Tanger-Tétouan-Al Hoceima (E)	312	160	+Domain	**0.792**	+27.9
Drâa-Tafilalet (D)	170	27	Baseline	0.812	−2.9
Drâa-Tafilalet (E)	170	86	+Domain	**0.735**	+22.9
Souss-Massa (D)	96	11	+Infra	0.885	0.0
Souss-Massa (E)	96	37	Baseline	**0.740**	+12.5
Marrakech-Safi (D)	69	8	+Infra	0.841	−4.3
Marrakech-Safi (E)	69	42	Baseline	**0.696**	+8.7
Eddakhla-Oued Eddahab (D)	34	8	+Infra	**0.824**	+5.9
Eddakhla-Oued Eddahab (E)	34	3	+Domain	0.882	−2.9

D = Difficult-vs-not, E = Easy-vs-not. Source:
`results/json/training/phase5_paper2_best_feature_results.json`.

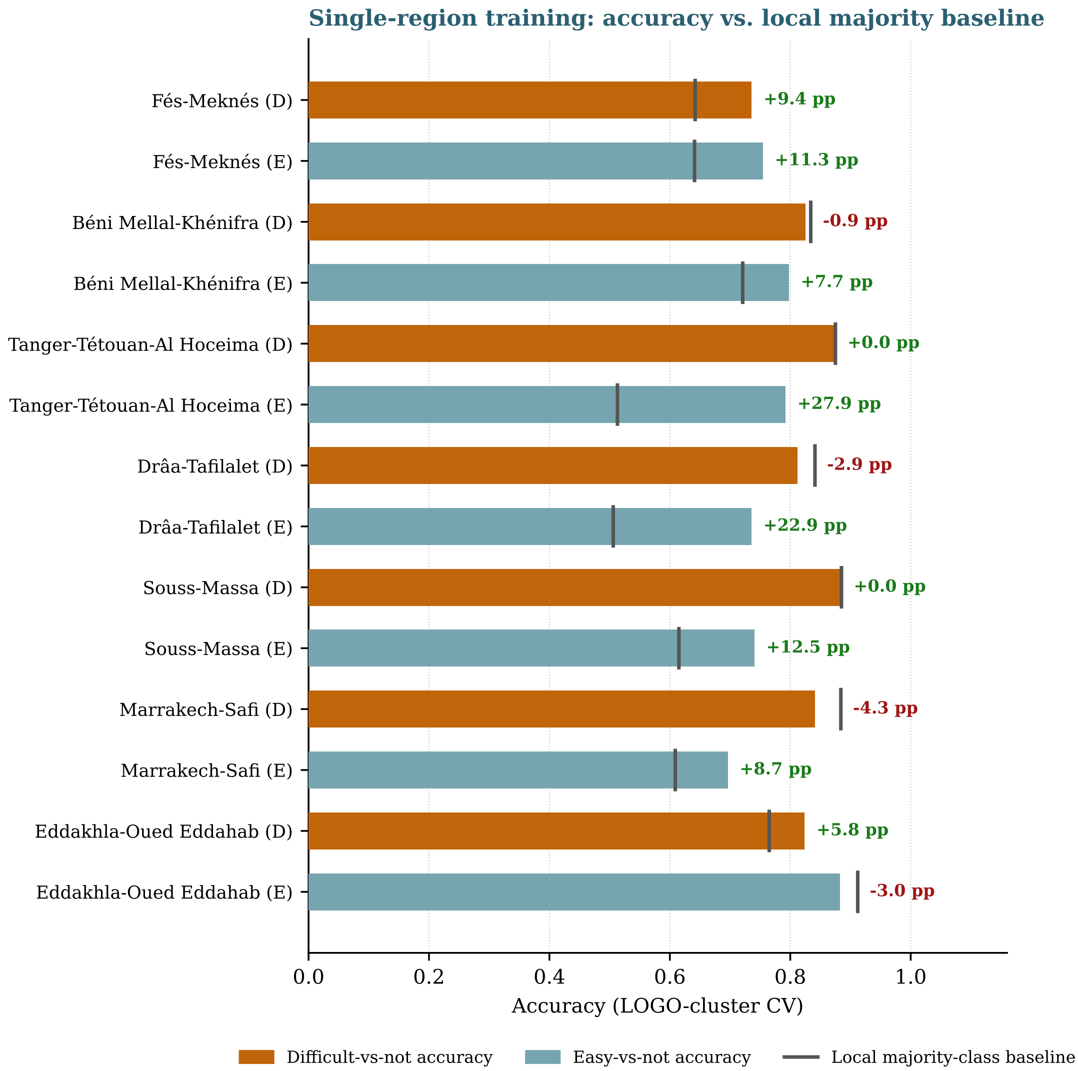

Figure 2. Single-region training: accuracy vs. local majority baseline, seven individually-modeled regions, both targets.

Eight of the fourteen region/target combinations in Table 4 beat their local majority baseline outright, two tie it exactly (Tanger-Tétouan-Al Hoceima Difficult, Souss-Massa Difficult), and four fall short even with the best available feature set (Béni Mellal-Khénifra Difficult, Drâa-Tafilalet Difficult, Marrakech-Safi Difficult, Eddakhla-Oued Eddahab Easy). Three of these four shortfalls are the Difficult target specifically, consistent with the mechanism traced in Section 6.

Why three regions are missing from Table 4 and Figure 2. Guelmim-Oued Noun, Laâyoune-Sakia El Hamra, and Casablanca-Settat are each modeled only as part of a merged group (Section 4.3); their results appear in the next subsection.

5.2 Merged/Multi-Region Results

Two merged groups are tested: Guelmim-Oued Noun with Laâyoune-Sakia El Hamra (and, as a further check, that pair with Eddakhla-Oued Eddahab added, to see whether the pair's result holds up at larger N), and Rabat-Salé-Kénitra with Casablanca-Settat. This is disclosed as a deliberate trade-off – a merged group is not the same object as a single administrative region – made specifically to get a usable result rather than an unusable one for the thinnest region in each pair. Table 5 and Figure 3 report these merged results the same way as the single-region ones above. Section 5.5 additionally resolves, for the final national map, which configuration (a single merged model, versus each sub-region modeled on its own) best represents each such zone.

Table 5. Merged/multi-region training: best result per merged group, both binary targets. Columns as in Table 4.

Region group	*N*	Pos.	Best feature set	Acc.	Gap (pp)
Guelmim + Laâyoune (D)	56	5	Baseline	0.857	−5.4
Guelmim + Laâyoune (E)	56	30	Baseline	**0.839**	+30.3
Guelmim + Laâyoune + Eddakhla (D)	90	13	+Infra	0.856	0.0
Guelmim + Laâyoune + Eddakhla (E)	90	33	Baseline	**0.878**	+24.5
Rabat-Salé-Kénitra + Casablanca-Settat (D)	58	6	+Domain	0.897	0.0
Rabat-Salé-Kénitra + Casablanca-Settat (E)	58	27	Baseline	**0.690**	+15.6

D = Difficult-vs-not, E = Easy-vs-not. Source:
`results/json/training/phase5_paper2_merged_regions_results.json`.

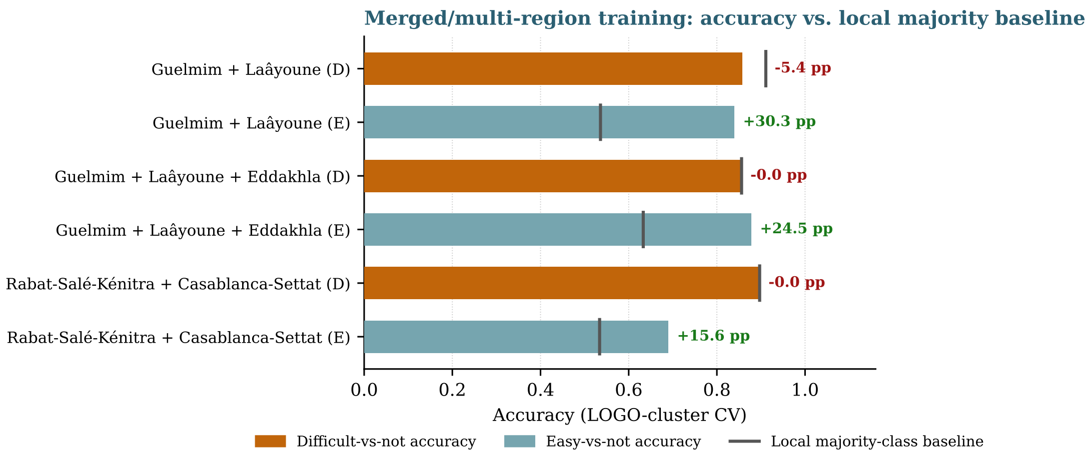

Figure 3. Merged/multi-region training: accuracy vs. local majority baseline, the three merged groups, both targets.

Three of the six merged-group combinations beat their local majority baseline, one by a very wide margin (Guelmim+Laâyoune Easy, +30.3 pp); two tie it exactly (the trio's own Difficult target, and Rabat+Casablanca Difficult); only Guelmim+Laâyoune Difficult falls short (−5.4 pp) – the Difficult target again. Combined with the single-region results, eleven of the twenty region/target combinations tested in this paper beat their local majority baseline outright, four tie it, and five fall short.

5.3 Region-by-Region Detail

Each region below is discussed on its own terms: what its labeled data looks like, which feature set won and why that is plausible given the region's terrain and infrastructure character, what the result does and does not establish, and its own projected map. Geological domain has no continuous spatial raster in this project (it is a curated per-locality label, not derived from any geology shapefile), so wherever +Domain wins, the *map* for that target is rendered from the Baseline feature set instead – disclosed explicitly per region below – while the accuracy numbers in Tables 4–5 and Figures 2–3 always reflect the true winning variant. This is a permanent limitation of the Domain feature, not a processing failure: unlike the +Infra feature set, whose regional maps all render correctly following a rewrite of the tourism-infrastructure extraction pipeline (Section 5.5), there is no equivalent fix available for Domain, since the underlying geological-domain layer simply does not exist as a continuous, grid-computable raster.

Fés-Meknés ($N = 537$**, largest region).** Spans the Fés-Boulemane plateau, Middle Atlas foothills, and lowland plains – the most heterogeneous single region in the dataset, and the historical source of a disproportionate share of the Difficult label (Section 6), meaning this region's own terrain-difficulty relationship is what a national pooled model learns best. Tourism-infrastructure features deliver a solid Difficult-class gain (+9.4 pp over baseline), plausible given the region's genuinely uneven tourism development between its plateau towns and remoter Middle Atlas approaches – exactly the kind of variation straight-line road distance alone cannot see. Geological domain wins Easy by a wider margin still (+11.3 pp). The Difficult map uses its true winning +Infra variant directly (no substitution); the Easy map substitutes Baseline for the true winning +Domain variant.

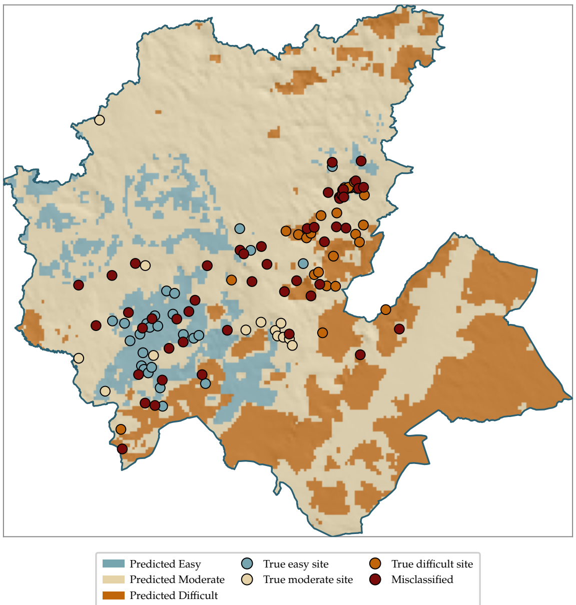

Figure 4. Fés-Meknés: projected map (Difficult: +Infra, true winner, acc. 0.736; Easy map shown with Baseline features, substituting for the true winning +Domain variant, acc. 0.754).

Béni Mellal-Khénifra ($N = 326$**).** Combines High Atlas relief with the Tadla plain. Even the best available feature set (the plain baseline) does not quite clear the local majority for Difficult (−0.9 pp) – consistent with this region's karst caves and river-canyon sites, several of which sit geographically close to roads yet require technical approach that none of the three feature sets tested can represent. The Easy classifier tells a different story: tourism-infrastructure features produce a large, genuine gain (+7.7 pp), the region's accessible Tadla-plain sites evidently correlating well with real nearby services. Both maps use their true winning variant directly, no substitution.

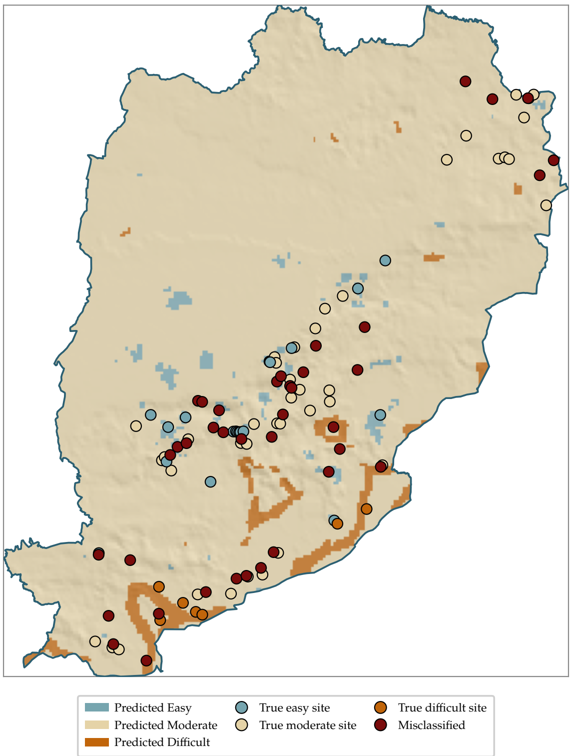

Figure 5. Béni Mellal-Khénifra: projected map (Difficult: Baseline, true winner though it falls just short of the local majority, acc. 0.825; Easy: +Infra, acc. 0.798).

Tanger-Tétouan-Al Hoceima ($N = 312$**, more than doubled since the previous labeling round: 128→312 sites).** Geological domain is the best feature set for both targets. The Difficult result ties the local majority baseline exactly (0.0 pp gap) – this region's substantial growth in labeled data did not, on its own, translate into a model that beats simply guessing the majority class, a finding worth flagging honestly rather than reporting the raw accuracy number without this context. Easy tells the opposite story: this is this paper's single largest individually-modeled-region gain (+27.9 pp) – a genuinely large, real improvement. Both targets' maps substitute Baseline for the true winning +Domain variant.

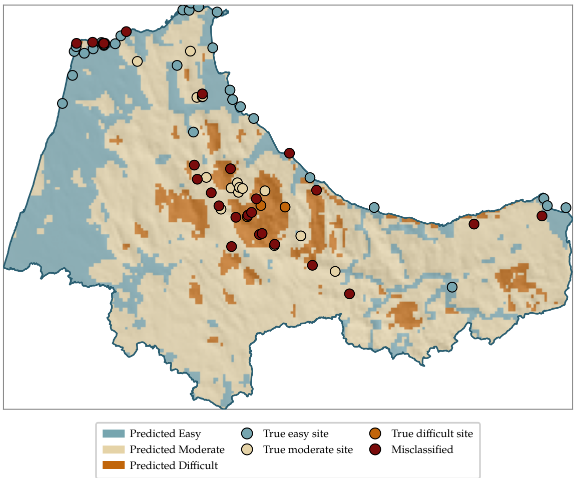

Figure 6. Tanger-Tétouan-Al Hoceima: projected map (both targets shown with Baseline features, substituting for the true winning +Domain variant: Difficult acc. 0.875; Easy acc. 0.792).

Drâa-Tafilalet ($N = 170$**, mixed environment).** The region used to test and reject the terrain-regime hypothesis (Section 6). The plain baseline wins Difficult but still falls just short of the local majority (−2.9 pp); geological domain wins Easy by a wide margin (+22.9 pp) – the clearest single-region illustration in this paper that the same region can see one target improve substantially while the other's underlying difficulty remains unresolved. The Difficult map uses its true winning Baseline variant directly; the Easy map substitutes Baseline for the true winning +Domain variant.

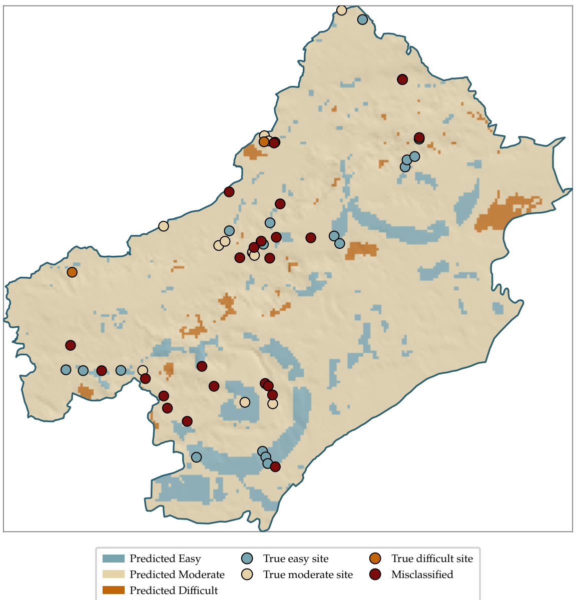

Figure 7. Drâa-Tafilalet: projected map (Difficult: Baseline, true winner, acc. 0.812; Easy map shown with Baseline features, substituting for the true winning +Domain variant, acc. 0.735).

Souss-Massa ($N = 96$**).** Tourism-infrastructure features win Difficult, exactly tying the local majority (thin positive class, $n = 11$); the plain baseline wins Easy with a real, solid gain (+12.5 pp). The Difficult map uses its true winning +Infra variant directly; the Easy map uses its true winning Baseline variant directly.

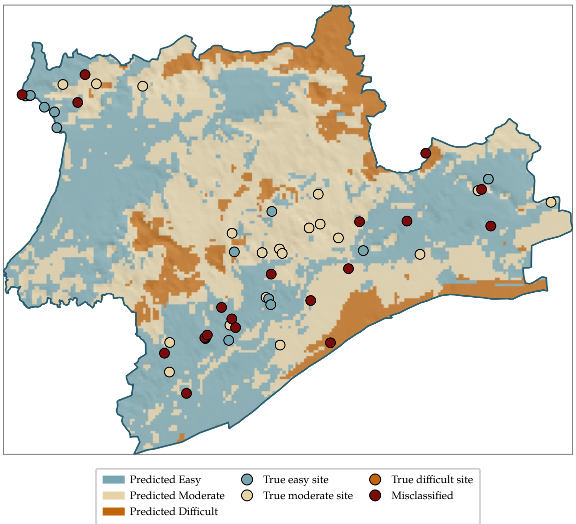

Figure 8. Souss-Massa: projected map (Difficult: +Infra, true winner, acc. 0.885; Easy: Baseline, true winner, acc. 0.740).

Marrakech-Safi (*N* = 69). The starkest split in this paper. Easy shows a modest, real gain (+8.7 pp, plain baseline, no feature engineering needed), while Difficult is this paper's single worst individually-modeled result even after trying all three feature sets (−4.3 pp – the best available option, tourism infrastructure, still falls short of simply guessing not-Difficult). With only 8 Difficult-labeled sites in this region, this result should be read as a region where the Difficult class specifically remains essentially unmodeled with current data and features, not as a settled finding. Both maps use their true winning variant directly, no substitution.

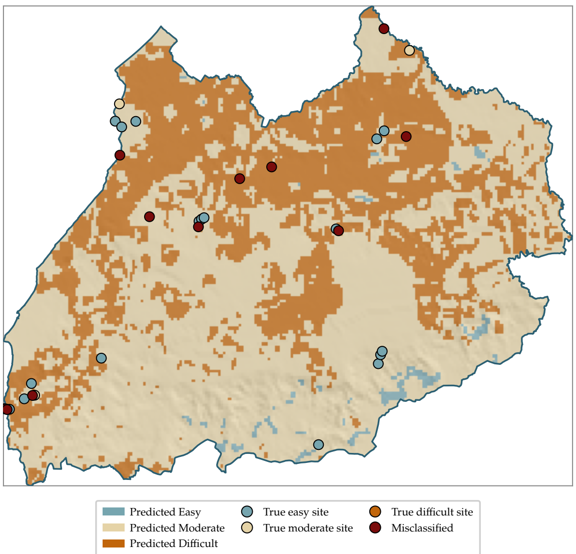

Figure 9. Marrakech-Safi: projected map (Difficult: +Infra, true winner though it falls short of the local majority, acc. 0.841; Easy: Baseline, true winner, acc. 0.696).

Eddakhla-Oued Eddahab (*N* = 34). Flat, arid, low-relief Saharan coastal terrain. Tourism infrastructure produces a real Difficult-class gain (+5.9 pp); Easy has only 3 positive examples in the entire region, too few for any feature set to move meaningfully beyond the baseline's near-majority result (−2.9 pp) – see also Section 5.4's LightGBM finding on this exact region. The Difficult map uses its true winning +Infra variant directly; the Easy map substitutes Baseline for the true winning +Domain variant. **This standalone result is shown here as its own analysis, but Section 5.5 finds that the merged trio configuration below represents this region better on the national map, on both targets** – the two results are not in conflict, they simply answer different questions (how does Eddakhla perform alone, versus what best represents Eddakhla's territory on a combined map).

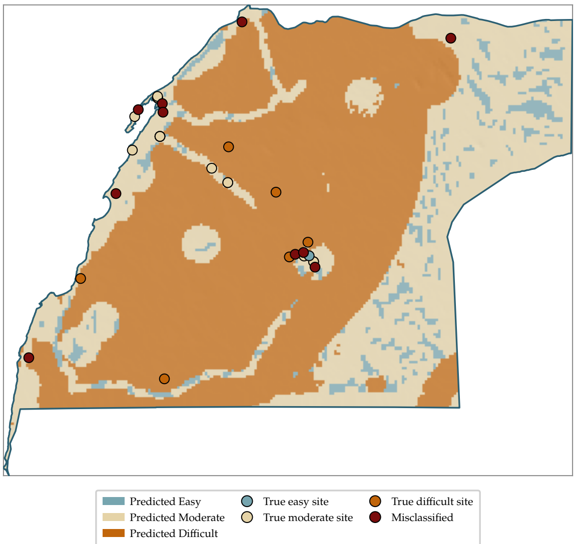

Figure 10. Eddakhla-Oued Eddahab (standalone): projected map (Difficult: +Infra, true winner, acc. 0.824; Easy map shown with Baseline features, substituting for the true winning +Domain variant, acc. 0.882). Superseded by the trio configuration for the national mosaic (Section 5.5).

Guelmim-Oued Noun + Laâyoune-Sakia El Hamra (N = 56, merged). The plain baseline wins both targets. Easy shows this entire paper's largest gain (+30.3 pp) – a result that did not exist in any usable form before merging, not a small improvement on an existing one. Difficult falls short of the local majority (−5.4 pp), the widest shortfall of any combination in this paper. Both maps use the true winning Baseline variant directly.

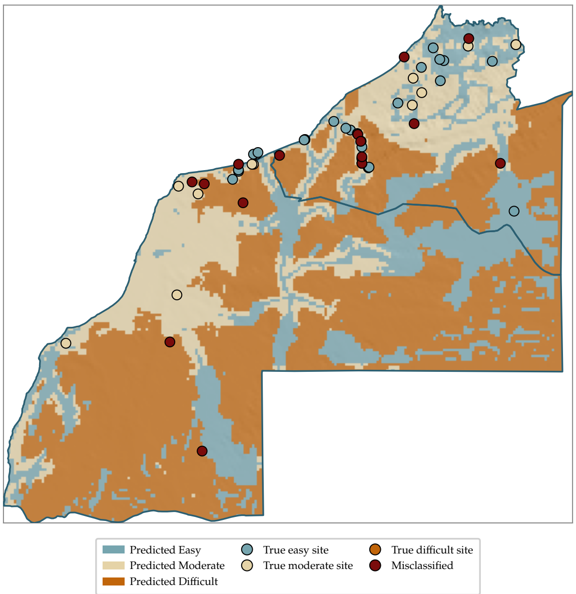

Figure 11. Guelmim-Oued Noun + Laâyoune-Sakia El Hamra (standalone duo): projected map (both targets: Baseline; Difficult acc. 0.857; Easy acc. 0.839). Superseded by the trio configuration for the national mosaic (Section 5.5).

Guelmim-Oued Noun + Laâyoune-Sakia El Hamra + Eddakhla-Oued Eddahab (N = 90, merged). Adding Eddakhla to the pair above increases the sample size to 90 and shows both targets still comparing favorably to their local majority (Difficult ties it exactly, Easy beats it by +24.5 pp). Unlike the earlier N=939-era version of this comparison, this trio configuration is not just a numeric side-check here: Section 5.5 shows it now genuinely outperforms modeling Eddakhla and the southern duo independently, on both targets, when compared fairly on Eddakhla's own sites – so it is the configuration actually used for the national mosaic's southern zone, not the standalone duo/Eddakhla maps shown above.

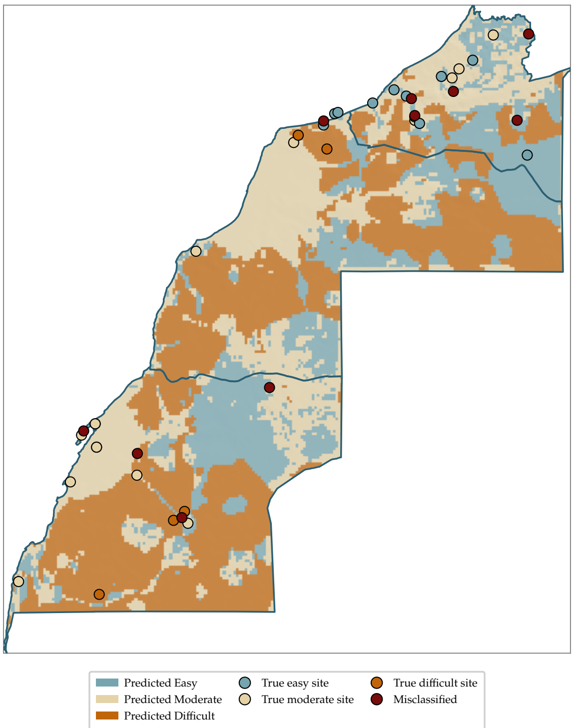

Figure 12. Guelmim-Oued Noun + Laâyoune-Sakia El Hamra + Eddakhla-Oued Eddahab (trio): projected map (Difficult: +Infra, acc. 0.856; Easy: Baseline, acc. 0.878). This is the configuration used for the southern zone of the national mosaic (Section 5.5).

Rabat-Salé-Kénitra + Casablanca-Settat ($N = 58$, merged). A more mixed result than the southern merge: geological domain lifts Difficult to an exact tie with the local majority (0.0 pp), while the plain baseline delivers a solid Easy-class gain (+15.6 pp) – broadly consistent with the Difficult/Easy asymmetry seen across most other regions in this paper. The Difficult map substitutes Baseline for the true winning +Domain variant; the Easy map uses its true winning Baseline variant directly.

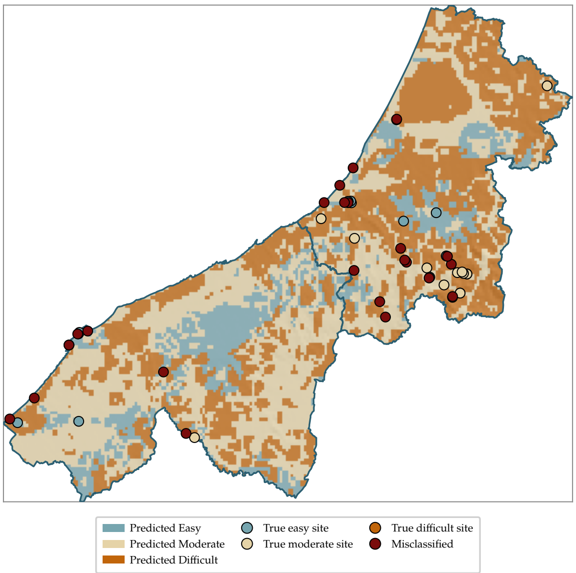

Figure 13. Rabat-Salé-Kénitra + Casablanca-Settat: projected map (Difficult map shown with Baseline features, substituting for the true winning +Domain variant, acc. 0.897; Easy: Baseline, true winner, acc. 0.690).

5.4 Per-Algorithm Comparison

Table 6 breaks out each region's four tree/boosting algorithms individually, on the six-feature baseline, rather than the soft-voted ensemble reported above – the explicit comparative view requested for this paper.

Table 6. Random Forest / XGBoost / Gradient Boosting / LightGBM, individually, per region and target (six-feature baseline, 500 m LOGO-cluster CV). Ensemble column is the four-model soft vote. **Bold** marks the best accuracy in each row.

Region (target)	RF	XGB	GBM	LightGBM	Ensemble
Fés-Meknés (D)	0.700	0.693	**0.702**	0.689	**0.702**
Fés-Meknés (E)	**0.747**	0.745	0.724	0.737	**0.749**
Béni Mellal-Khénifra (D)	0.794	0.819	0.807	**0.837**	0.825
Béni Mellal-Khénifra (E)	**0.733**	**0.733**	0.715	0.724	0.715
Tanger-Tétouan-Al Hoceima (D)	0.856	0.830	**0.869**	0.853	0.862
Tanger-Tétouan-Al Hoceima (E)	**0.788**	0.756	0.779	0.760	0.785
Drâa-Tafilalet (D)	0.788	0.788	**0.818**	**0.818**	0.812
Drâa-Tafilalet (E)	0.676	0.706	0.700	0.688	**0.718**
Souss-Massa (D)	0.833	0.833	0.812	0.823	**0.854**
Souss-Massa (E)	0.667	**0.708**	0.656	0.594	**0.740**
Marrakech-Safi (D)	**0.841**	0.783	0.826	0.768	0.797
Marrakech-Safi (E)	0.667	0.609	0.580	0.580	**0.696**
Eddakhla-Oued Eddahab (D)	0.824	0.794	0.735	**1.000**	0.765
Eddakhla-Oued Eddahab (E)	0.853	**0.882**	**0.882**	0.794	0.853

D = Difficult-vs-not, E = Easy-vs-not. Source: `results/json/training/phase5_per_algorithm_regional_results.json`.

Individual algorithms vary by up to roughly 11 percentage points within the same region and target (Souss-Massa Easy: XGBoost 0.708 vs. LightGBM 0.594) – smaller than, and not as systematic as, the variance driven by feature-set choice (Table 4) or by which region is being modeled at all.

The bolded LightGBM= 1.000 entry (Eddakhla-Oued Eddahab, Difficult, $N = 34$, only 8 positive examples) is a genuine result of this fixed, deterministic pipeline, not a scripting error: Random Forest (0.824) and XGBoost (0.794) already beat this region's local majority baseline (0.765) on the same 34-site split, and Gradient Boosting (0.735) lands close to it, so a perfect score from a fourth algorithm on the same small, flat, low-relief, low-noise region is a difference of degree, not a contradiction of the other three. A plausible reading is that with only 8 positive examples in a genuinely simple terrain setting, the true decision boundary may be close to perfectly separable by a small number of clean splits, which LightGBM's specific configuration happened to find here and the other three did not quite reach. Tellingly, the four-model soft-voted ensemble reported elsewhere in this paper (0.765) lands *below* every individual algorithm except Gradient Boosting – averaging a perfect learner in with three imperfect ones pulls the combined vote down rather than up, a concrete illustration that an ensemble's headline number does not automatically inherit its best member's performance.

5.5 National Mosaic Map

Figure 14 assembles the per-region and per-merged-group maps of Section 5.3 into a single national-extent map, each zone shaded by its own best-performing model rather than one pooled national model, with regional administrative borders overlaid.

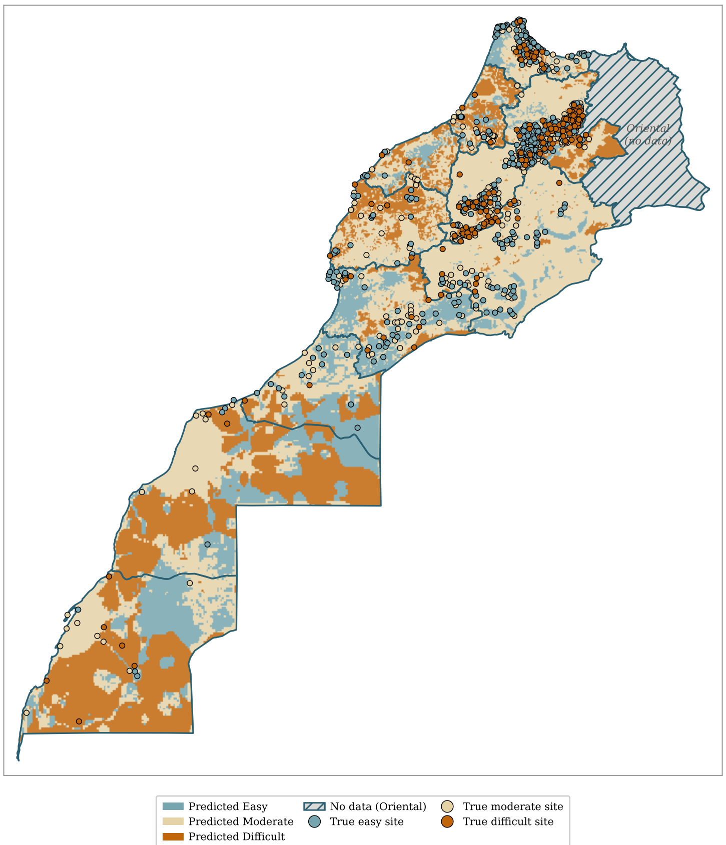

Figure 14. National mosaic: eight modeled zones covering eleven of Morocco's twelve regions, each shaded by its own best-performing region-specific model. Oriental (grey, hatched) carries only 4 labeled geosites, still too few to model or merge meaningfully, and is shown as missing data, not filled from a national fallback model.

Zone-composition decisions. Two of the eight modeled zones had more than one candidate configuration available (Section 4), and the choice for each was resolved quantitatively, per target, rather than by default. Both comparisons below were re-run for this update after finding and fixing a bug that had corrupted the original N=939-era comparison: the refitting script had used a hardcoded 0.5 decision threshold rather than each configuration's own tuned threshold, which silently understated some configurations' true accuracy.

Southern zone (Guelmim-Oued Noun, Laâyoune-Sakia El Hamra, Eddakhla-Oued Eddahab). Comparing Eddakhla's own standalone model against the full trio's model, restricted to Eddakhla's own 34 sites (the fair, apples-to-apples test, since the trio is trained on more data but only this subset is common ground): the trio wins both targets – Difficult 0.853 vs. 0.824 standalone, Easy 0.912 vs. 0.882 standalone. (Before the threshold fix, the Easy trio figure was wrongly computed as 0.765, making standalone look better; the corrected, properly-thresholded number reverses that conclusion.) The trio configuration is therefore used for the whole southern zone on the national mosaic, on both targets – a change from how this comparison was resolved before this update, when the two were computed as an exact tie. The southern duo's and Eddakhla's own standalone maps (Section 5.3) remain valid, separate analyses of what each smaller unit achieves on its own; they are simply not the configuration used for the mosaic.

Rabat-Salé-Kénitra + Casablanca-Settat. Rabat-Salé-Kénitra has grown enough (Difficult $n_{\text{pos}} = 4$) to support a real standalone comparison for the first time – previously it had too few Difficult-labeled sites once separated from Casablanca-Settat to fit at all. Restricted to Rabat's own 44 sites: Difficult is an exact tie (0.909 both ways), and Easy favors standalone Rabat by +4.55 pp (0.750 vs. 0.705 merged). Despite Rabat's own preference for standalone modeling, Casablanca-Settat itself still has no standalone fit of its own ($N = 14$, the thinnest region in the catalog) and so cannot be shown on the map on its own – the merged configuration is therefore kept for the whole zone, as the only one with full coverage of both territories, at a small, disclosed accuracy cost on Rabat's own Easy target and no cost on Difficult (Source: `02_modeling_and_analysis/28_paper2_rabatcasa_zone_compare.py`, `results/json/other/phase5_paper2_rabatcasa_zone_comparison.json`).

Coverage and limitations. The mosaic covers eleven of Morocco's twelve regions and 1,658 of the 1,662 labeled sites in the catalog (the 4 Oriental sites are excluded, per the decision above). Oriental is shown as missing data rather than modeled from any national or neighboring-region fallback, per an explicit decision to disclose the gap honestly rather than paper over it with an unvalidated extrapolation: even with 4 labeled sites now, none of them Difficult, there still is not enough ground truth to check any model covering this region against. A second, previously disclosed limitation no longer applies to this version of the mosaic: earlier renderings of these maps substituted Baseline features for some +Infra-winning targets because the region-scale tourism-infrastructure extraction failed under a memory cap; that extraction pipeline has since been rewritten (a streaming, node-by-node approach replacing the previous GeoDataFrame-based one) and now completes reliably nationwide, so every +Infra-winning target's map genuinely uses +Infra features. The only remaining, permanent substitution is for +Domain-winning targets (Section 5.3 discloses each individually), since geological domain has no continuous spatial raster to draw a map from; the underlying accuracy numbers in Tables 4–5 always reflect the true winning variant regardless of what the map shows.

6 Robustness and Limitations

This section reports findings from a deliberate, systematic robustness investigation – not incidental observations, but the direct product of testing, and where warranted ruling out, candidate explanations for why the Difficult class specifically underperforms across most regions. All numbers below are recomputed fresh at $N = 1{,}662$; two findings changed in a way worth reporting honestly rather than silently updating.

The dominant, evidenced cause: one region's terrain signature standing in for the whole labeled Difficult class – now confirmed a third time. The three labeling batches making up this dataset are not evenly spread across Fés-Meknés: the original batch is 44.0% Fés-Meknés by count, the second (2026 expert) batch only 7.3%, and the third (2026-08-28) batch 27.6%. A model trained on this pool learns, in large part, the Fés-Meknés-specific relationship between terrain/road proximity and difficulty (a mountain-terrain, moderate-road-distance signature) rather than a general one. Testing this directly, using each batch's own predictions from the national baseline model: Difficult precision is 0.540 in the Fés-Meknés-heavy original batch, but only 0.324 in the second batch (which barely touches Fés-Meknés) and 0.362 in the third batch (which touches it only moderately) – three independently sourced batches, with three different Fés-Meknés shares, showing precision that tracks that share almost exactly. This rules out labeler-specific error as the cause (three different labeling efforts show the same pattern) and confirms a base-rate/threshold-calibration mechanism: a single global decision threshold, implicitly tuned toward Fés-Meknés' unusually high local Difficult rate, over-predicts Difficult wherever the true local rate is much lower, producing exactly the false-positive-heavy precision collapse observed in the batches that are not Fés-Meknés-heavy.

A terrain-regime explanation was tested and specifically ruled out, more clearly than before. Before settling on the prevalence mechanism above, an elevation/road-distance "regime split" hypothesis (mountain-near-road versus remote-lowland difficulty as two distinct terrain mechanisms) was tested directly, splitting all Difficult sites into two regimes using fixed elevation and highway-distance thresholds. Measured properly with precision rather than raw accuracy, the remote-lowland regime performs clearly better than the mountain regime (precision 0.550 vs. 0.366, a wider gap than found in the previous version of this test) – the opposite of what the terrain-regime hypothesis predicted. This candidate explanation is reported here specifically because it was refuted by evidence, not because it confirmed anything, consistent with reporting negative results rather than only positive ones.

A previously promising routing-distance feature no longer reaches statistical confirmation at this larger sample size. Road-network routing distance (true shortest-path distance along the OSM road graph, in place of straight-line highway distance) was tested under leave-region-out evaluation (training on all other regions, zero exposure to the target region) for the two regions with the strongest theoretical interest, Tanger-Tétouan-Al Hoceima and Souss-Massa. At $N = 1{,}662$, the raw per-region gains are real but noticeably smaller than before (Tanger-Tétouan-Al Hoceima +1.6 pp, Souss-Massa +5.2 pp, versus +7.0 pp and +11.9 pp previously), and the pooled comparison across the two regions is now only borderline ($\chi^2 = 2.89$, $p = 0.089$) rather than clearly significant. Repeating the same test across every region with a usable leave-region-out sample remains clearly non-significant ($\chi^2 = 0.26$, $p = 0.611$), essentially unchanged from before. The honest reading, updated from the previous version of this finding: road-network routing distance may still be a small, regionally-selective improvement in a couple of regions, but the evidence for it is weaker at this larger sample size, not stronger, and it should not be reported as a confirmed regional fix.

Algorithm choice is a smaller factor than feature-set and regional context. Section 5.4's per-algorithm table shows individual tree/boosting algorithms varying by up to roughly 11 percentage points within the same region and target (excluding the one extreme, explained outlier discussed there), but this variance is smaller and less systematic than the variance driven by feature-set choice (Table 4) or by which region is being modeled at all – consistent with the national companion paper's model-family comparison, where switching the underlying algorithm family (tree ensemble vs. k-nearest-neighbors vs. Gaussian Process) moved national accuracy by several percentage points in either direction depending on the target, a similar order of magnitude to what is seen here regionally.

Hyperparameter selection is not fully nested within the outer cross-validation. As in the national companion paper: for each reported model, the best configuration among Random Forest/XGBoost/Gradient Boosting/LightGBM (and, here, the best of three feature-set variants) is chosen once via grid search on the full regional dataset, then reused across every fold of the outer leave-one-group-out evaluation that produces the headline accuracy. This is a common simplification in applied work, not full nested cross-validation, disclosed here rather than treated as fully addressed – and is a larger concern at the region-specific scale of this paper than nationally, since smaller N per region gives model selection relatively more of the total sample to have already seen.

7 Conclusion

Applying the same feature stack and validation protocol used nationally, region by region, with each region allowed its own best-performing feature set among three candidates, shows a genuinely uneven picture at $N = 1{,}662$: eleven of twenty region/target combinations beat their local majority-class baseline outright, four tie it exactly, and five fall short even with the best available feature set – concentrated, not coincidentally, in the Difficult class. The largest gains are substantial (Tanger-Tétouan-Al Hoceima Easy +27.9 pp, the merged Guelmim-Oued Noun/Laâyoune-Sakia El Hamra group +30.3 pp on the same target), while the weakest results (Marrakech-Safi Difficult, the same southern group's own Difficult target) show that a handful of region/target combinations remain essentially unmodeled with current data and features, not a settled negative finding but an honest limit of what this approach currently achieves there. The recurring Difficult-class weakness is traced to a specific, now three-times-independently-confirmed mechanism rather than left as an unexplained weakness: one region's terrain signature (Fés-Meknés) implicitly sets the model's decision threshold, which then over-predicts Difficult wherever the true local rate is much lower – and a competing terrain-regime explanation was tested and ruled out along the way, more clearly than in the previous version of this analysis. A previously reported routing-distance improvement is honestly downgraded here: it no longer reaches clear statistical confirmation at this larger sample size, even in the two regions where it looked most promising before. Three regions, still the thinnest in the catalog even after this update, are merged with a geographic neighbor to become usable, and a corrected zone-by-zone comparison – fixing a real threshold bug found during this update – newly determines that the southern trio configuration, not the previously assumed independent split, best represents that zone on the final map. Every region's and merged group's result is projected onto a real geographic grid and shown as its own map, combined into a single national mosaic covering eleven of Morocco's twelve regions – the clearest visual summary this project has produced of where region-specific accessibility modeling currently works, where it does not, and why. Together with the per-algorithm comparison (Section 5.4), this paper provides the detailed, region-by-region performance/robustness/limitations picture that the national companion paper's pooled figures are not designed to show, and sets up the natural next step: a focused, single-region deep dive selected from the strongest and most instructive results reported here.